

No. of Functional Features Included in the Solution

Date	03 July 2026
Team ID	XXXXXX
Project Name	Smart Lender – Loan Eligibility Prediction Using Machine Learning
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features implemented in the **Smart Lender – Loan Eligibility Prediction Using Machine Learning** project. Each feature is described with its purpose, implementation status, module/component, and contribution to the overall solution. The application provides an end-to-end workflow from applicant data entry to real-time loan eligibility prediction using a trained **XGBoost** machine learning model integrated with a **Flask** web application.

S.No	Feature Name	Feature Description	Module/ Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Allows users to enter applicant details such as gender, marital status, education, self-employed status, applicant income, co-applicant income, loan amount, loan amount term, credit history, property area, and dependents	Frontend (HTML5, CSS3, JavaScript)	Done	High
2	Input Validation	Validates mandatory fields and user inputs before processing the loan eligibility prediction.	Flask Backend	Done	High
3	Data Preprocessing	Handles missing values, encodes categorical features, scales numerical values, and prepares input data for prediction.	Preprocessing Module	Done	High
4	Loan Eligibility Prediction	Predicts whether the loan application should be approved or rejected using the trained XGBoost machine learning model.	Machine Learning Model	Done	High
5	Result Display	Displays the prediction result (Approved/Rejected) immediately after processing the applicant information.	Flask Application	Done	Medium

6	Trained Model Loading	Loads the trained machine learning model (model.pkl) and preprocessing scaler (scaler.pkl) during application startup.	Model Loader	Done	Medium
7	Responsive User Interface	Provides a responsive and user-friendly web interface accessible from desktop and mobile browsers.	HTML5, CSS3, JavaScript	Done	Medium
8	Error Handling	Handles invalid, incomplete, or incorrect user inputs gracefully using validation and exception handling.	Flask Backend	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (User Input Form, Input Validation, Data Preprocessing, Loan Eligibility Prediction, Result Display)
Additional / Nice-to-Have Features	3 (Trained Model Loading, Responsive User Interface, Error Handling)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	User Input Form, Result Display
2	Backend / Logic	3	Input Validation, Data Preprocessing, Loan Eligibility Prediction
3	Database / Storage	1	CSV Dataset, model.pkl, scaler.pkl
4	API / Integration	1	Flask Application Routes
5	Security / Authentication	1	Input Validation and Basic Error Handling